



[Original Paper]

Application of big data in smart agriculture

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Abstract

Using big data in smart agriculture represents a significant technological revolution in the agricultural sector. This paper delves into the primary application areas of big data in smart agriculture, encompassing precision agriculture, intelligent farming machinery, and sustainable agriculture. To begin with, big data technology offers comprehensive data support for decision-makers in agriculture. Through real-time monitoring and extensive data analysis, agricultural producers can gain more precise insights into soil conditions, crop growth status, and other crucial information. Furthermore, implementing big data in precision agriculture facilitates intelligent agricultural production by integrating precision fertilization, irrigation, and pest and disease monitoring, thereby enhancing production efficiency. The advancement of intelligent farming machinery also plays a pivotal role in enhancing the intelligence and efficiency of field operations. Moreover, big data technology contributes to sustainable agriculture by providing scientific support for agricultural decisions through data analysis on climate change, resource utilization, and other pertinent aspects, thereby promoting the transition towards environmentally friendly agricultural practices. Looking ahead, with continuous innovation and widespread adoption of big data technology, smart agriculture is poised for significant development opportunities.

Keywords

big data in agriculture, smart agriculture technology, precision agriculture applications, intelligent farming machinery, sustainable agriculture with big data

1. Introduction

The rapid evolution of big data technology is ushering in profound changes in the field of agriculture. In recent years, due to the continuous advancement of information technology, the volume of data generated in agricultural production has expanded rapidly. This phenomenon, commonly referred to as “big data,” has emerged as a driving force behind the modernization of agriculture. The significance of big data in smart agriculture is increasingly prominent, acting as a crucial support for enhancing agricultural production efficiency and optimizing resource utilization. By adeptly collecting, managing, and analyzing this extensive agricultural data, decision-makers in agriculture can gain more precise insights into various factors such as soil conditions, weather patterns, crop growth, and market information. This, in turn, enables more intelligent agricultural production ^[1-5].

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The application of big data in smart agriculture first faces the challenge of data collection and management. The data involved in agricultural production covers a wide range, including soil composition, weather changes, and plant growth conditions. Various technologies such as sensors and satellite remote sensing can be employed to monitor and collect this data in real time. Simultaneously, for such massive data, scientific management and storage become the foundation for subsequent analysis. Therefore, constructing an efficient big data collection and management system not only provides data support for smart agriculture but also lays the groundwork for more in-depth decision analysis [6-10].

The application of big data in smart agriculture goes beyond data collection; it crucially involves transforming this data into concrete decision support. Data cleaning, processing, and analysis become indispensable steps. Through big data analytics, crop diseases can be identified, optimal planting times can be determined, and market demand can be predicted, providing agricultural decision-makers with more scientific and precise information support. The real-time and comprehensive nature of big data allows agricultural decisions to be adjusted in real-time, minimizing production risks and enhancing agricultural production efficiency. Therefore, the role of big data in decision support is considered one of the core drivers of smart agriculture [11-15].

This paper aims to delve into the application of big data in smart agriculture, with a focus on analyzing the impact of big data technology on various aspects of agricultural production and emphasizing its importance for the development of smart agriculture. By comprehensively introducing specific applications of big data in data collection, management, decision support, and more, the goal is to provide readers with a profound understanding of big data technology in smart agriculture.

The structure of this paper is as follows. Starting from the introduction, the rise of big data in agriculture and its importance to smart agriculture are presented. Subsequently, the paper is divided into two key sections. The first part focuses on the collection and management of big data, providing an in-depth analysis of data sources and collection technologies in agricultural production, emphasizing the necessity of constructing an efficient big data collection and management system. The second part concentrates on the role of big data in decision support, covering data cleaning, processing, analysis, and highlighting the crucial support provided by big data to decision-making in smart agriculture. Through discussions on these two main aspects, readers will gain a comprehensive understanding of the key applications of big data in smart agriculture, thus better recognizing its role in improving agricultural production efficiency and sustainable development

2. Data collection and management in big data agriculture

2.1. Data sources in agricultural production

The sources of data in agricultural production are diverse and extensive, covering various aspects from land to weather and plants. Firstly, soil composition is a crucial data source in agriculture. Through soil sensors and detection devices, agricultural producers can obtain key information about soil nutrient content,

and pH values, aiding in the formulation of scientific fertilization plans. Secondly, meteorological data plays a vital role in agriculture. Weather stations and satellite technology can provide accurate meteorological information, including temperature, humidity, and rainfall, offering real-time weather conditions to assist farmers in planning planting and agricultural activities. Additionally, plant growth conditions are also an essential data source in agriculture. Through image sensors and monitoring devices, practitioners can monitor crop growth in real-time, facilitating timely measures against pests and diseases.

Among these data sources, agricultural producers can gain in-depth insights into critical elements such as land, weather, and plants, enabling better production planning and decision-making. The high-quality collection of this data forms the foundation of smart agriculture, laying the groundwork for subsequent big data analysis and applications.

The diversity of data sources allows agricultural producers to comprehensively understand the conditions of farmland, and scientifically formulate planting plans, and agricultural schedules. By fully utilizing modern sensors and monitoring technologies, agricultural producers can not only increase production efficiency but also achieve precise monitoring of the agricultural production environment. This not only helps improve crop yield and quality but also contributes to reducing resource waste and environmental pollution. Therefore, a deep understanding of data sources in agricultural production promotes the development of smart agriculture, achieving a more sustainable and efficient agricultural production model.

2.2. Big data collection technologies

The successful application of big data in smart agriculture relies on efficient big data collection technologies. Firstly, sensor technology plays a crucial role in agricultural data collection. Various sensors, such as soil sensors, weather sensors, and image sensors, can monitor and collect multidimensional data in real-time from the agricultural production environment. The widespread use of these sensors enables agricultural producers to accurately obtain key information, such as soil nutrient content, weather conditions, and plant growth status, providing real-time and comprehensive data support for smart agriculture. Secondly, satellite remote sensing technology is also a significant means of big data collection. Through satellite remote sensing, agricultural producers can obtain land information over a broader area, including land use conditions and vegetation indices. This remote monitoring technology provides a more comprehensive and efficient means of data collection for larger-scale farmland. Additionally, advanced agricultural machinery is equipped with various sensors, enabling automatic monitoring of the agricultural production process, and facilitating the real-time generation of data.

With the support of big data collection technologies, the acquisition of agricultural data no longer relies on traditional manual collection or single-channel data sources but is achieved through automated and intelligent means, enabling large-scale, real-time, and comprehensive data collection.

Building an efficient big data collection and management system is a prerequisite for the implementation

of smart agriculture. This requires agricultural producers to integrate various data sources reasonably, ensuring data quality and consistency. Simultaneously, there are higher demands for the real-time processing and transmission of data to ensure the timely and effective application of this data. Therefore, in the development of big data collection technologies, attention should be paid not only to the means of data collection but also to the establishment of an efficient data management system to support more in-depth agricultural data analysis and applications.

3. Big data analysis techniques

3.1. Data cleaning and processing

In the application of big data in smart agriculture, data cleaning and processing are crucial steps to ensure data quality and usability. Firstly, data cleaning involves filtering, removing outliers, and filling in missing values in the collected raw data to ensure accuracy and completeness. In agricultural data, environmental factors and limitations of sensor equipment may introduce noise or outliers, and data cleaning effectively enhances the credibility of the data. Secondly, data processing involves manipulating, organizing, and calculating cleaned data to obtain higher-level information. In smart agriculture, data processing includes using statistical methods, machine learning algorithms, etc., to extract valuable information from massive data, such as crop growth trends, and pest predictions, providing scientific support for agricultural decision-making.

The effectiveness of data cleaning and processing is directly related to the reliability of subsequent data analysis and applications. Through standardized data cleaning and processing procedures, practitioners can obtain more accurate and reliable data, thereby enhancing the scientific and operational aspects of decision-making. Additionally, effective data cleaning and processing provide a high-quality data foundation for the visualization and model building of agricultural big data. Therefore, data cleaning and processing play a vital role in smart agriculture, laying a solid foundation for the in-depth application of big data.

3.2. Data mining and model building

In smart agriculture, data mining and model building are critical processes for transforming vast agricultural data into actionable information. Firstly, data mining involves using various algorithms and techniques to discover potential patterns and rules from extensive agricultural data. This includes operations like classification, clustering, and association rule mining to extract useful information. For instance, through data mining, agricultural practitioners can discover the optimal timing for crop growth under specific meteorological conditions or identify potential factors affecting yields. Secondly, model building is the construction of mathematical or statistical models based on data mining to better understand and predict the behavior of agricultural systems. Machine learning algorithms play a crucial role in this process by learning from historical data to establish predictive models, aiding agricultural decision-makers in making

scientifically sound decisions.

The goal of data mining and model building is to reveal patterns and associations hidden in agricultural data through in-depth analysis, providing more accurate guidance for agricultural production. By establishing predictive models, practitioners can better address risks such as climate change and pest outbreaks, achieving precise management of agricultural production. Additionally, by continuously optimizing and adjusting models, they can adapt to different environments and conditions, enhancing the accuracy of predictions. Therefore, data mining and model building provide robust technical support for the intelligence and sustainable development of smart agriculture.

3.3. Application of machine learning in agriculture

The application of machine learning technology in smart agriculture adds a new dimension to agricultural decision-making. Firstly, through supervised learning algorithms, machine learning can utilize extensive historical data to train models, predicting future agricultural trends and production conditions. For example, machine learning models can be used to forecast crop yields and the probability of pest occurrences, providing more accurate planting recommendations for agricultural producers. Secondly, unsupervised learning algorithms play a crucial role in clustering and classification tasks in agriculture. Through clustering analysis of different plots, plants, or soil types, machine learning helps agricultural practitioners better understand the heterogeneity of land, providing a basis for precision management. Additionally, the introduction of reinforcement learning algorithms makes autonomous and intelligent agricultural machinery possible. Through the learning and adaptability of intelligent farm machinery, more efficient field operations can be achieved.

The widespread application of machine learning in agriculture not only improves the efficiency of agricultural production but also injects new vitality into the sustainable development of agriculture. By continuously optimizing and adjusting models, machine learning technology can adapt to complex environments such as different weather, land, and plant conditions, providing more personalized and accurate solutions for agricultural decision-making. This allows agricultural production to better cope with challenges such as climate change and resource limitations, optimizing resource utilization and driving agriculture toward a more intelligent and sustainable direction. Therefore, the application of machine learning in agriculture not only enhances the level of modernization in agriculture but also brings new possibilities for the sustainable development of the agricultural industry.

4. Role of big data in decision support for agriculture

4.1. Demands for agricultural production decision-making

The demands for decision-making in agricultural production constitute one of the core driving forces behind the development of smart agriculture. Firstly, with the continuous increase in the global population,

the demand for food is on the rise. To meet this demand, agricultural production must become more efficient and intelligent. Agricultural producers need to make accurate planting decisions under constantly changing weather conditions, scientifically allocate resources, and enhance crop yield and quality. Secondly, the intensification of climate change imposes higher requirements on agricultural production. Irregular rainfall, extreme weather events, and other factors disrupt traditional agricultural seasons and cycles, necessitating more flexible adjustments to planting plans and irrigation schemes to adapt to the evolving weather conditions. Additionally, intensified global market competition requires agricultural producers to finely adjust the planting structure, engaging in intelligent agricultural production based on market demand.

These challenges and demands drive agricultural producers to seek more scientific and intelligent decision support. The development of smart agriculture aims to meet the urgent need of agricultural producers to make scientifically informed decisions by integrating advanced technologies in the face of constantly changing environments. Smart agriculture not only focuses on improving production efficiency but also emphasizes ensuring food supply within the framework of sustainable development, enhancing the adaptability and resilience of agricultural production. Therefore, the demand for decision-making in agricultural production propels the innovation and application of smart agricultural technologies, providing robust momentum for the transformation of the agricultural industry.

In summary, the demands for decision-making in agricultural production are increasingly complex and challenging, compelling agricultural producers to continuously seek more advanced technological means. Through the introduction of smart agricultural technologies, agricultural producers can better cope with uncertainties such as climate change and market fluctuations, achieving intelligent and sustainable development of agricultural production. Thus, the demand for decision-making in agricultural production is not only the driving force behind the development of smart agriculture but also an urgent requirement for achieving modernization and sustainable development in agriculture.

4.2. Practical applications of big data in agricultural decision-making

The practical applications of big data in agricultural decision-making have yielded various achievements, primarily evident in precision agriculture. Through large-scale data collection and analysis, agricultural producers can obtain more detailed information, such as precise soil details, meteorological data, and plant growth conditions, enabling precise fertilization and irrigation. This not only improves farmland utilization efficiency, and reduces waste, but also helps enhance the quality and yield of agricultural products. Secondly, significant progress has been made in the application of big data in agricultural production planning. By analyzing historical yields, market demands, and other data, agricultural decision-makers can more accurately predict future market trends, adjust crop planting structures, and implement more scientifically informed agricultural production plans. Big data plays a crucial role in pest and disease monitoring and control, allowing decision-makers to detect and respond to outbreaks earlier, ensuring the stability of

agricultural production.

In agricultural decision-making, big data also provides robust support for agricultural finance. By analyzing various data on agricultural production, financial institutions can more comprehensively assess the operational status of farms, offering more personalized financial services to agricultural producers. Additionally, big data has extensive applications in logistics and supply chain management. Through real-time monitoring and analysis of logistics data, the transportation, storage, and sale of agricultural products can be more efficiently carried out, reducing losses and transportation costs, and improving the efficiency of product circulation.

Overall, the practical applications of big data in agricultural decision-making have profoundly changed the way agricultural production operates. From precision agriculture to market forecasting, from pest and disease monitoring to financial services, big data technology provides decision-makers in agriculture with more comprehensive and accurate information support in different domains. This application not only enhances the efficiency of agricultural production but also promotes the modernization and sustainable development of the agricultural industry. Therefore, the practical application of big data in agricultural decision-making provides agricultural producers with more innovative possibilities, driving the digital transformation of the agriculture sector.

5. Application of big data in precision agriculture

5.1. Utilization of satellite remote sensing data

The widespread application of satellite remote sensing data provides crucial information support for agricultural decision-making. Firstly, satellite remote sensing data can offer extensive, high-resolution surface information in agriculture. Through data captured by satellite sensors, agricultural practitioners can obtain detailed information about land use, vegetation growth, soil moisture, and more. These data not only provide a comprehensive spatial perspective for agricultural production but also assist producers in better understanding and monitoring changes in different land plots. Secondly, time-series analysis of satellite remote sensing data can track agricultural seasonal changes. By comparing satellite data at different time points, agricultural decision-makers can understand the dynamic process of plant growth, predict crop growth cycles, identify potential issues promptly, and take corresponding measures. This is crucial for planting plans, irrigation management, and pest monitoring.

The utilization of satellite remote sensing data plays a key role in precision agriculture. Through the comprehensive analysis of multi-source satellite data, agricultural decision-makers can formulate more intelligent and scientific agricultural strategies. For example, the health of vegetation can be assessed based on vegetation indices, allowing timely adjustments to fertilization plans. Monitoring soil moisture enables precise irrigation management. Moreover, satellite remote sensing data can be used to monitor the impact of natural disasters such as floods and droughts, enabling proactive measures against agricultural disasters.

Therefore, the utilization of satellite remote sensing data not only provides timely and comprehensive information for agricultural decision-making but also plays an irreplaceable role in improving agricultural production efficiency and reducing disaster risks.

5.2. Role of drone technology in big data

Drone technology plays an increasingly crucial role in big data applications, especially in the field of agriculture. Firstly, drones, equipped with various sensors and cameras, can efficiently and rapidly collect various data in agricultural production. For instance, infrared cameras can be used to monitor crop growth status, and multispectral sensors can provide useful information such as vegetation indices. This high-frequency, high-resolution data collection allows agricultural decision-makers to have a more comprehensive understanding of farmland conditions, and promptly identify issues like plant diseases or nutrient deficiencies. Secondly, drone technology achieves precise monitoring of large-scale farmland. Through automatic flight and remote sensing technology, drones can cover vast agricultural areas in a short period, providing detailed geographic information. This offers an efficient means for agricultural practitioners to better formulate planting plans, and irrigation schemes, enhancing the sustainability and profitability of agricultural production.

The role of drone technology in big data goes beyond data collection; it extends to supporting big data analysis and decision-making. By incorporating advanced image processing and computing capabilities, drones can process and analyze collected data in real-time, providing feedback to agricultural practitioners. This provides powerful technical support for the development of precision and smart agriculture. Additionally, drone technology plays a role in agricultural supervision, enabling precise monitoring of farmland area, and crop planting conditions, contributing to accurate agricultural production supervision. Therefore, in the era of big data, drone technology provides agriculture with more efficient, precise data support, propelling agriculture towards digitization and intelligence.

5.3. Application of big data in precision fertilization and irrigation

The application of big data in precision fertilization and irrigation has brought revolutionary changes to modern agriculture. Firstly, through big data analysis of soil nutrient levels, vegetation indices, meteorological conditions, and other multidimensional data, agricultural decision-makers can formulate more precise fertilization plans. Big data technology can monitor soil nutrient levels in real-time, analyze crop growth status, providing scientifically sound fertilization recommendations. This not only improves the efficiency of fertilization, reduces resource waste in agricultural production but also helps ensure the quality of agricultural products. Secondly, the application of big data in irrigation management is equally significant. Through monitoring soil moisture, meteorological conditions, and other data, big data technology can intelligently adjust irrigation systems, achieving precise plant irrigation. This precision irrigation not only

improves the efficiency of water resource utilization but also helps prevent issues like soil salinization, promoting sustainable development in agriculture.

The application of big data in precision fertilization and irrigation not only enhances the efficiency of agricultural production but also has a positive impact on the environment and sustainable agriculture. By accurately understanding the needs of soil and plants, agricultural decision-makers can reduce unnecessary use of fertilizers and water resources, lowering the environmental impact of agriculture. This lays the foundation for the green development of agriculture. Additionally, the application of big data in precision fertilization and irrigation provides a practical case for agricultural digitization, driving the development of smart agriculture. Through the integration of big data technology, agricultural production gradually transitions from experience-driven to data-driven and science-driven, providing more scientific and reliable support for agricultural decision-making.

6. Conclusions

The application of big data in smart agriculture has already brought profound impacts, providing vast prospects for the future development of the agricultural sector. Firstly, big data technology has offered a fresh perspective and tools for agricultural production. Through real-time monitoring and extensive data analysis, agricultural decision-makers can gain a more comprehensive and accurate understanding of critical information such as farmland conditions and crop growth status, enabling them to formulate more scientifically sound decisions. The application of big data not only enhances the efficiency of agricultural production but also serves as an opportunity for the digital transformation of the agricultural industry.

Secondly, the influence of big data on smart agriculture manifests in various aspects, including precision agriculture, smart agricultural machinery, and sustainable agriculture. With the support of big data, agricultural producers can achieve precision fertilization, irrigation, and pest monitoring, thereby increasing production efficiency. The development of smart agricultural machinery also contributes to more intelligent and efficient field operations. Additionally, big data provides crucial information support for the sustainable development of agriculture, assisting decision-makers in better addressing challenges such as climate change and resource limitations.

In conclusion, looking ahead to the future, with the continuous development and widespread adoption of big data technology, smart agriculture will usher in even broader development prospects. Future smart agriculture will emphasize characteristics such as digitization, intelligence, and sustainability. Big data technology will continue to drive scientific and intelligent decision-making in agriculture, enhancing the adaptability and resilience of agricultural production. Concurrently, addressing challenges like data security, quality, and technology accessibility will be essential to promote the healthy development of smart agriculture. In the future, big data will continue to bring more innovation to agriculture, propelling the agricultural industry toward greater modernization and sustainability.

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